Krzysztof Suberlak

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U.S. Citizen (naturalized)

SUMMARY

Research scientist with 6+ years at the Vera C. Rubin Observatory developing image analysis algorithms, physics-based predictive models, and machine learning pipelines on large, multi-source scientific data. Core expertise in wavefront estimation from defocused imagery, Bayesian inference, time-series modeling, and comparative uncertainty quantification across physics-based and learned models. Co-author on AIDonut, a production PyTorch CNN deployed in the Rubin pipeline. Multi-source data integration at terabyte scale across six large astronomical catalogs. Field-deployed through three on-site commissioning campaigns in Chile. Prior NASA satellite remote-sensing experience. Fluent in Python; PhD in Astronomy.

EXPERIENCE

Research Scientist / **Engineer** Jan 2025 – Present
Vera C. Rubin Observatory Active Optics System (AOS) | DiRAC Institute, University of Washington, Seattle, WA

- Lead algorithm and pipeline development for the closed-loop optical calibration subsystem of the 8.4 m Simonyi Survey Telescope with its 3.2-gigapixel LSSTCam: a multi-year research-to-production effort combining physics-based forward models, deep-learning inversion, and uncertainty quantification. Build and maintain Python packages (`ts_wep`, `donut_viz`, `ts_aos_analysis`) deployed to production within the LSST Science Pipelines, following standard engineering practices (Git/GitHub, continuous integration, peer code review, unit testing).
- Developed image analysis and inverse-problem algorithms that recover the 3D optical state of the telescope from paired 2D defocal detector images at four wavefront-sensor locations across the LSSTCam focal plane — combining the physics-based Transport of Intensity Equation, forward-model fitting (GalSim) in the Zernike polynomial basis, and uncertainty propagation across the pipeline.
- Deployed and benchmarked **AIDonut**: a PyTorch-based deep-learning model for the same inverse problem (Crenshaw et al. 2024, AJ 167:86; co-author) at scale on the LSST US Data Facility (USDF) alongside baseline physics-based algorithms, enabling comparative accuracy studies and uncertainty-aware model selection.
- Provide on-sky remote support for LSSTCam/LSSTComCam night operations and maintain the interactive dashboards (deployed to **RubinTV**, the observatory's public monitoring interface) that track pipeline throughput, model performance, and image-quality metrics; used live during observing to inform algorithm-selection decisions, sometimes with after-hours coverage.
- Author technical notes (e.g., SITCOMTN-156 on image-pair count and binning effects on optical calibration accuracy); present to internal stakeholders and the community (LSST@Europe7, Poznań, September 2025; Rubin Project & Community Workshops).

Postdoctoral Scholar Dec 2019 – Jan 2025
Vera C. Rubin Observatory Commissioning, AOS Team | DiRAC Institute, University of Washington, Seattle, WA

- Maintained and extended the physics-based wavefront inversion pipeline: source detection in defocused imagery, image alignment, Zernike coefficient estimation, and closed-loop convergence monitoring; achieved $\sim$100$\times$ speed-up in the object-detection step.
- Built physics-based simulation infrastructure (PhoSim, imSim) to generate labelled training and validation data with known ground truth: essential for verifying inversion accuracy independent of the convergence loop.

- Developed and maintained core Python packages (`ts_wep`, `ts_phosim`, `ts_imsim`, `donut_viz`); contributed pull requests and resolved tickets across multiple LSST Science Pipelines packages.
- Developed quantitative data-quality metrics (SNR, entropy, centroid-offset) and diagnosed root causes of systematic issues (coordinate systems, detector gain and read-noise calibration); implemented solutions in the production pipelines.
- Multi-source data integration across six large sky surveys (LSST, SDSS, Pan-STARRS1, ZTF, DECaPS, GAIA DR2) at terabyte scale using SQL, Apache Spark, and AXS at the USDF; resolved batch-processing, memory, and data-access-layer constraints.
- Participated hands-on in three on-site commissioning campaigns at Cerro Pachón, Chile (2023 integration; December 2024 ComCam; May 2025 LsstCam): real-time analysis and decision support during remote field deployments at a mountain-top observatory.
- Concurrent research on stochastic time-series modeling using Gaussian processes and continuous-time autoregressive models (CARMA / Damped Random Walk) applied to multi-source quasar light curves; two first-author publications (ApJ, MNRAS).
- Mentored four student research projects on multi-survey time-series analysis and multi-source data fusion (2021–2024).

EARLIER SATELLITE REMOTE-SENSING EXPERIENCE

Cassini CIRS Spacecraft Instrument Data Analysis (NASA/ESA Cassini mission) Autumn 2012
Atmospheric, Oceanic and Planetary Physics, University of Oxford, UK

Analyzed thermal-infrared spectral data from the Cassini Composite Infrared Spectrometer (CIRS) using NEMESIS radiative-transfer retrieval; derived vertical atmospheric structure of Jupiter (NH_3, PH_3, hydrocarbons, aerosols) via chi-squared fitting with propagated retrieval uncertainties.

LRO Diviner Spacecraft Instrument Support (NASA Lunar Reconnaissance Orbiter) Summer 2010
Atmospheric, Oceanic and Planetary Physics, University of Oxford, UK

Laboratory calibration measurements for the Diviner thermal-radiometer aboard NASA's Lunar Reconnaissance Orbiter: NIR/FIR spectroscopy of lunar-analog materials under vacuum + cryogenic conditions; characterized quartz grain-size distributions via image analysis; analyzed on-orbit Diviner data in IDL.

EDUCATION

PhD, Astronomy — University of Washington, Seattle, WA 2019
Thesis: *Quasar Variability as seen by Large Optical Sky Surveys.* Advisor: Prof. Željko Ivezić.

MPhys, Physics — University of Oxford, United Kingdom 2012

TECHNICAL SKILLS

Statistical Modeling, ML, and Computer Vision: Bayesian inference (MCMC); Gaussian processes and continuous-time autoregressive modeling (CARMA / Damped Random Walk); uncertainty propagation across physics-based and learned models; classical ML (random forests); deep learning via PyTorch (co-author of AIDonut, a production CNN).

Image Processing: Image analysis algorithm development for CCD sensors and reflective telescopes; image-pair registration and comparative analysis; wavefront estimation from defocused imagery; image-quality metrics (SNR, entropy, centroid-offset); sensor calibration and detector systematics (gain, read-noise, vignetting, blending, dark/bias correction); instrument signature removal; production deployment.

Inverse Problems and Physics-Based Modeling: Transport of Intensity Equation; forward-model fitting with GalSim in Zernike polynomial basis; radiative-transfer retrievals (NEMESIS); physics-based synthetic data generation for ML training and verification (PhoSim, imSim).

Multi-Source Data Integration and ETL: Multi-survey catalog integration at TB scale across six large sky

surveys (LSST, SDSS, Pan-STARRS1, ZTF, DECaPS, GAIA DR2); SQL (PostgreSQL-backed), Apache Spark, AXS, LSST Butler (data-access layer); schema reconciliation and feature engineering across heterogeneous data sources.

Distributed Computing: BPS (Batch Production System), HTCondor, Slurm; pipelines deployed at scale on the LSST US Data Facility (USDF).

Languages: Python (fluent; NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch), SQL, IDL, Bash, LaTeX; C++ (undergraduate coursework, modern-C++ self-study).

Software Engineering: Git/GitHub, code review, continuous integration (Jenkins), unit testing; agile tools (Jira, Confluence).

Visualization and Operations: Interactive dashboards (RubinTV); Grafana, Chronograf; real-time telemetry analysis; on-sky observing support during commissioning.

Geospatial Foundations (in progress): Astronomical world coordinate systems (WCS), spherical geometry, catalog spatial cross-matching; actively building Earth-geospatial toolchain fluency (GDAL, GeoPandas, rasterio) through hands-on PyTorch semantic-segmentation and object-detection projects on open satellite imagery.

SELECTED PUBLICATIONS

- **Suberlak, K.L.**, Ivezić, Ž., MacLeod, C.L. "Improving Damped Random Walk Parameters for SDSS Stripe 82 Quasars with Pan-STARRS1." *The Astrophysical Journal* 907:96 (2021).
- Crenshaw, J.F., Connolly, A.J., Meyers, J.E., Kalmbach, J.B., Megias Homar, G., Ribeiro, T., **Suberlak, K.L.**, Thomas, S., Tsai, T.-W. "Using AI for Wave-front Estimation with the Rubin Observatory Active Optics System." *The Astronomical Journal* 167:86 (2024).
- **Suberlak, K.L.**, Ivezić, Ž., MacLeod, C.L., Graham, M., Sesar, B. "Solving the Puzzle of Discrepant Quasar Variability on Monthly Time-scales Implied by SDSS and CRTS Data Sets." *MNRAS* 472(4):4870 (2017).

SELECTED TECHNICAL NOTES

- SITCOMTN-156 (2025): ComCam — impact of image-pair count and binning settings on optical calibration accuracy.
- SITCOMTN-146, -144 (2024): curvature-wavefront-sensing analyses of on-sky commissioning data (LsstComCam, LsstCam).
- SITCOMTN-139, -108, -085, -072, -046, -044, -038 (2020–2024): AOS algorithm, wavefront estimation, and simulation studies.

PROFESSIONAL SERVICE

Peer reviewer: *The Astrophysical Journal* (2 manuscripts); *Monthly Notices of the Royal Astronomical Society* (2 manuscripts). Mentored four student research projects on multi-survey time-series and data-fusion analysis.

HONORS AND AWARDS

- UW eScience Data Science for Social Good Fellow (2015).
- DIRAC Institute Fellow, University of Washington (2016 – present).
- UW Data Science Postdoctoral Fellowship (2023).
- UW Jacobsen Award (2019).
- American Astronomical Society Travel Award (2016).
- Royal Astronomical Society Travel Grant (2016).
- Fellow of the Royal Astronomical Society (2008 – present).